

Andrew Koja

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Education

Michigan State University

Bachelor of Science in Computer Science

Aug 2024 – May 2028

East Lansing, MI

- **Relevant Coursework:** CSE 231: Programming I (Python), CSE 232: Programming II (C++)

Experience

Apostles4Jesus

Full-Stack Software Developer

June 2023 – March 2026

Macomb, MI

- Designed and deployed a bilingual (English/Arabic) digital podcast platform for an organization with a combined 15,000+ social media following using HTML/CSS, JS, and Netlify
- Mitigated OWASP Top 10 security risks by implementing CSP, X-Frame-Options, rate limiting, and reCAPTCHA via serverless Netlify Functions
- Implemented persistent localization preferences, hreflang SEO optimization, and Google Analytics tracking for a multilingual UX

Projects

LightAI | React, FastAPI, PyTorch, SQLite, WebSockets, APScheduler

lightai-kohl.vercel.app

- Engineered an API monitoring platform with per-endpoint anomaly detection, processing 28,800+ daily latency readings under 100ms
- Developed a root cause analysis engine correlating endpoint failures and linking latency regressions to GitHub deployments
- Shipped a pip-installable Python SDK featuring a non-blocking `@monitor` decorator for async function instrumentation

CrimeScope | React, Vite, Leaflet, Zustand, Recharts, SODA / ArcGIS APIs

crime-scope.vercel.app

- Normalized data from 17 U.S. city APIs, enabling client-side filtering of 50,000+ records without a backend
- Rendered an interactive visualization dashboard featuring pin, cluster, and heatmap mapping modes
- Implemented shareable URL state, stale-data detection, and city-aware date handling across 17 open-data portals

FootyLive | Python, FastAPI, React, TypeScript, SQLite, WebSockets, Redis

FootyLive

- Architected a real-time soccer dashboard delivering live scores for 6 European leagues every 30 seconds via WebSockets
- Developed custom reconciliation logic for UEFA Champions League knockout aggregation across two-leg ties
- Maintained uptime through automated background synchronization, Redis caching, and graceful SQLite fallback

FaceRead | React, Vite, face-api.js, MediaPipe, Web Audio API, WebGL

face-read.vercel.app

- Deployed face-api.js fully client-side via WebGL to run 3 neural networks in parallel at 15–30 FPS with zero server lag
- Built a browser-based ASL fingerspelling classifier using MediaPipe Hands keypoints supporting all 26 letters
- Reduced inference latency from 500ms to real-time by migrating the pipeline from a Flask backend to client-side WebGL

SharpEdge | React, Vite, Flask, SQLite, The Odds API

SharpEdge

- Constructed a full-stack odds comparison platform for major sports leagues, optimizing demand-driven API polling
- Resolved a data integrity flaw from timestamp drift across 9 sportsbooks by restructuring SQLite aggregations
- Visualized sharp money signals and line movements using historical odds snapshots on a custom comparison grid

Technical Skills

Languages & Frameworks: Python, JavaScript, TypeScript, Java, C++, HTML/CSS, SQL, Flask, FastAPI, React, Vite, Node.js

Tools & DevOps: Git/GitHub, Docker, SQLite, Redis, PyTorch, WebSockets, SQLAlchemy

Concepts: Machine Learning, Real-Time Systems, Anomaly Detection, REST APIs, Data Visualization, Server-Side Caching